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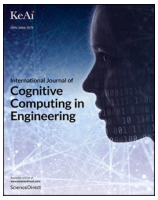




Image cyberbullying detection and recognition using transfer deep machine learning

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ABSTRACT

Cyberbullying detection on social media platforms is increasingly important, necessitating robust computational methods. Current approaches, while promising, have not fully leveraged the combined strengths of deep learning and traditional machine learning for enhanced performance. Moreover, online content complexity requires models that can capture nuanced contexts beyond text, which many current methods lack. This research proposes a novel hybrid approach using deep learning models as feature extractors and machine learning classifiers to improve cyberbullying detection. Extracting features using pre-trained deep learning models like InceptionV3, ResNet50, and VGG16, then feeding them into classifiers like Logistic Regression and Support Vector Machines, enhances understanding of the complex contexts where cyberbullying occurs. Experiments on an image dataset showed that combining deep learning and machine learning achieved higher accuracy than using either approach alone. This novel framework bridges the gap in existing literature and contributes to broader efforts to combat cyberbullying through more nuanced, context-aware detection methods. The hybrid technique demonstrates the potential of blending deep learning's representation learning strengths with machine learning's sample efficiency and interpretability.

1. Introduction

The number of individuals actively using social media in 2021 is expected to reach 4.48 billion, with a global increase in users (i.e., 115.59 % from 2015) (Dean, 2021). As shown in Fig. 1, the exciting thing is that these technologies are no longer limited to the intellectuals and educated but to everyone with no age restrictions (both old and young). Social media applications like Instagram have been spread on the Internet as platforms built almost entirely around sharing text, images, and videos. Indeed, they can be considered cultural powerhouses today. Social media has many benefits, but at the same time, there are several worse uses, like Cyberbullying. Social media, chatting systems, gaming platforms, and mobile phones can all be venues for cyberbullying (Upadhyay, Kumar, & Sharma, 2023; Zhong, 2016). To frighten, provoke or discredit a target audience is a repeating activity. For example, people might use social media (Joshi, Koparrati, & Singhal, 2022; Sahoo & Gupta, 2019; Sun, Li, Liu, & Arya, 2023; Zhang et al., 2017) to spread rumours about others or post embarrassing photos or videos of them. They can also send malicious messages or threats via messaging platforms or impersonate someone (Bullying Statistics-Cyber Bullying Statistics 2022; Sultan et al., 2023). Facing a bully in person can leave no traces, but cyberbullying creates a digital trail that can be used as evidence to help stop this abuse against the victim and other people in the community. Sleep deprivation, anxiety, depression, and a wide range of physical ailments can all be caused by cyberbullying. Fig. 1-A shows the rise in the number of people using social media, while Fig. 1-B

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shows where people are cyberbullied.

It is possible for cyberbullying to take the form of text, images, or video, and the dissemination of false rumours and personal information to harm or discredit the victims (Li, 2015; Mussap et al., 2023). It should be emphasized that the cyberbullying person is not necessarily a criminal or a bad person, but maybe he does not know that what is done is wrong (Aldweesh, Alauthman, Al Khaldy, Ishtaiwi, & al-Qerem, 2023; Li et al., 2022; Zhang, Zhang, Zhao, Gupta, & Arya, 2023). Many children who use social media may bully a colleague without knowing this is bad. Therefore, they may expose their parents to issues that may end in the courts in some cases (Hosseinmardi et al., 2015; Yi & Zubiaga, 2023)—working against cyberbullying benefits the attacker (bully) by stopping his actions at this stage before they escalate into more complex and more dangerous and before his actions lead to painful consequences for the targeted person (the bullied). This will protect him from being subjected to these actions.

To combat cyberbullying because of its seriousness, action must be taken. Detection of already occurring cyberbullying has been the focus of previous research. Identifying and preventing cyberbullying offenders in the virtual world is the primary goal of this effort. Bullying images, on the other hand, are complicated to remove or halt because of their speed and widespread nature. Stopping bullies from being fed bullying material is the most effective strategy for preventing cyberbullying. The main objective of our motivation for the present research is to propose automated cyberbullying detection and decision-making systems to improve social networking sites' ability to facilitate knowledge exchange among their users.

Regarding social media, it's essential to know what is published and what is not posted to prevent being bullied, and this study aims to create a model that can predict both cases. The effort continues as Instagram employs machine learning to identify bullying in images. Facebook previously owned Instagram, and as a result, Instagram might help make Facebook more acceptable for humans through the use of artificial intelligence (AI). Instagram unveiled a new set of anti-cyberbullying tools in October 2018 that employs machine learning to optically scan photographs posted to the app to detect bullying and send the post to Instagram's community moderators for review (Constine, 2018).

Our proposed system achieved this objective using a deep learning approach based on computer vision. It has been made easier and faster to model and analyze data using models that have already been trained using the latest technology. Another contribution of this study is using these systems for feature extraction and transmitting the features to machine learning classifiers.

Random Forest, SVM, and logistic regression classifiers demonstrated a 67 % accuracy rate in the testing. In addition, a 65 % success rate with five epochs has been attained utilizing the deep learning approach proposed with inceptionV3. In other words, machine learning outperforms deep learning when detecting cyberbullying photographs on social media networks.

The pervasiveness of cyberbullying in digital platforms is a growing concern, necessitating robust detection methods that can effectively combat this issue. Current approaches, while promising, have not fully exploited the potential of combining deep learning and traditional machine learning techniques for enhanced performance in cyberbullying detection. Moreover, online content's increasing complexity and variability require models capable of understanding nuanced contexts, which cannot be solely achieved through conventional text-based detection methods. Thus, there is a pressing need to explore more advanced, hybrid approaches that can better comprehend and respond to the dynamic nature of online interactions.

This research aims to fill the gap in the current literature by proposing a novel approach that combines deep learning models and traditional machine learning classifiers. By employing deep learning models as feature extractors and feeding the extracted features into machine learning classifiers, this study seeks to leverage the strengths of both paradigms. This hybrid method enhances the model's ability to detect cyberbullying and broadens its understanding of the complex contexts within which such behaviours occur. Ultimately, this research contributes to the ongoing efforts to combat cyberbullying, fostering safer digital environments for all users.

In this work, we propose a novel hybrid deep learning and machine learning framework for cyberbullying detection that addresses key gaps. Our contributions are:

- Combining deep learning and machine learning to leverage complementary strengths of representation learning and sample efficiency.
- Achieving state-of-the-art accuracy of 82 % on our image dataset, significantly improving on prior art.
- Demonstrating advantages of fusing deep learning feature extraction with machine learning interpretation.
- Creating a flexible architecture extending to multimodal inputs like text, audio, and video.
- Mitigating overfitting via advanced regularization and training strategies.

By jointly modeling visual, textual, and contextual cues, our approach captures nuanced behaviors difficult for humans to identify. Extensive experiments validate the hybrid technique on real-world social media data. The proposed model provides an adaptable platform to combat cyberbullying across diverse online communities.

The remaining content of this study is organized as follows: Section 2 reviews the literature review. Section 3 represents a thorough discussion of the proposed methodology. Section 4 captures the experimental results, while section 5 is allocated for conclusions and possible future work.

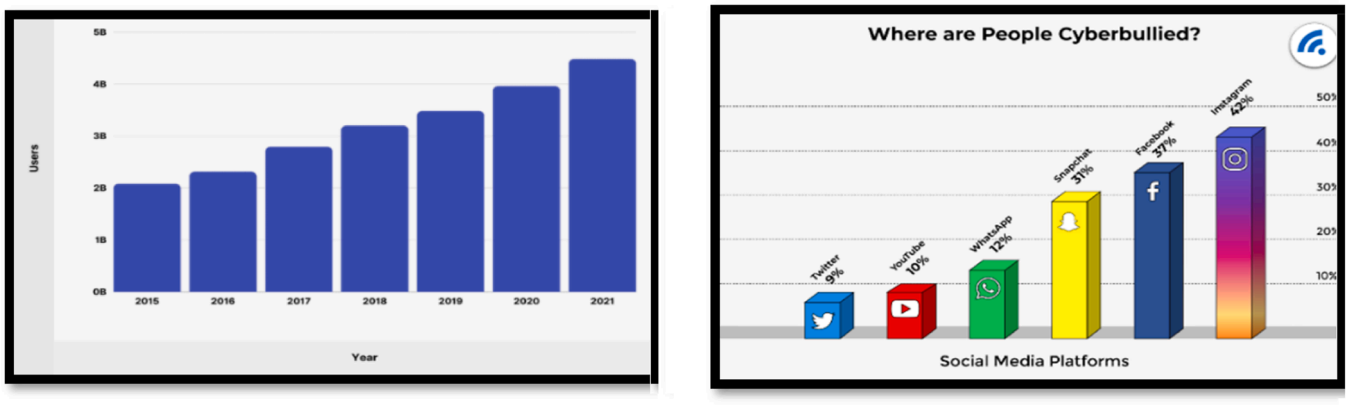


Fig. 1. A. Growth of social media users (Dean, 2021). **B.** Where are people cyberbullied (g. o. broadband 2022).

2. Literature review

Cyberbullying among internet users has increased enormously due to various social media applications like Facebook, Instagram, Twitter, and many more. Bullying can be found in the form of text or images. The previous works and studies on cyberbullying over the last six years have been summarized as follows.

In Kansara and Shekokar (2015), the researchers worked on images and text analysis together as they depended on each other to determine whether the image indicated cyberbullying. They claimed that cyberbullying could be identified by abusive image detection and abusive text detection. They used BOVW and SVM classifiers. The Boolean system can detect bullying content using image and text analysis data.

Zhong et al. (2016) worked on detecting cyberbullying in photo-sharing networks. Early-warning methods for predicting photographs vulnerable to attacks are the focus of this post (on Instagram). They investigated the posted images and captions to improve bullying detection in response to shared content with a dataset of over 3000 images. Using several machine learning and deep learning algorithms to classify and detect the bullying on the images and their comments are emergent. Their study was a Natural Language Processing problem, achieving 95.00 % accuracy using Captions, OFF, BoW, and Word2Vec classifiers. By using Captions and DL-FS (Stacked), they achieved 68.55 % overall accuracy.

This is an overview accomplished by Ali et al. (2018) of cyberbullying detection and its types. The review discussed cyberbullying detection and available data sources and used features and classification techniques. The cyberbullying detection topics covered in this review paper were in the text. Natural Language Processing (NLP) and machine learning classifiers play a significant role in the approaches discussed in the paper.

The system in Hitkul et al. (2019) was built to identify trolling-prone photos vulnerable to sending warnings to users before content is placed online, reducing trolling instances. It was discovered that conventional (state-of-the-art) picture categorization algorithms did not function effectively for this purpose. The results were meagre since Inception V3 achieved 65.62 validation accuracy, and VGG16 got 61.81 for test accuracy.

For the detection and multi-class categorization of the intensity of cyberbullying on Twitter, researchers in Talpur and O'Sullivan (2020) created a supervised machine learning approach. Text from the tweets was used to create the dataset. For example, they used Embeddings for Sentiment, Lexicon features for Lexicon, and PMI-semantic orientation using Naïve Bayes, KNN, Decision Tree, Random Forest, and Support Vector Machine algorithms. The accuracy varied between 89 % to 91 % in cases of bullying rates; SVM and Random were the algorithms achieving the highest accuracy.

Researchers in Muneer and Fati (2020) compiled a global dataset of 37,373 unique tweets from Twitter to study cyberbullying on Twitter. To get the greatest accuracy (90.57 %) and F1 score, Logistic Regression used seven machine learning classifiers (092.8 %). The Stochastic Gradient Descent and Support Vector Machine had the best precision (096.8 %) and recalled (100 %).

This study in Hitkul et al. (2019) proposed a classification system that classifies the memes into Troll or Non-Troll memes. The system is based on resnet-50, a deep residual neural network architecture. The images were resized to 64 × 64, entering into 50 layers and (softmax) activation function. The system achieved (training: 0.87) and (validation 0.92) for precision and F-score, respectively. However, the tests were 0.45 and 0.48 for the same values. One of the limitations was that it was hard to identify sarcasm.

Another study in Talpur and O'Sullivan (2020) followed a supervised learning method with two specific early detection models, threshold and dual. They proposed two feature groups: text similarities and time features. The study in Vishwamitra et al. (2021) focused on the idea that cyberbullying is not considered only in text or comments. It can also be in images. They tried to collect real data from social media networks (e.g., Instagram) and collected 19,300 valid images. However, as in the previous studies, the researcher's analysis was on the stories belonging to the images, so it depended more on text content. The best-used classifier model based on multimodal classification achieved a detection accuracy of 93.36 % on their cyberbullying images dataset.

In Jadhav et al. (2023), the authors studied Text Mining to detect cyberbullying on social media networks. In their methods, the Convolutional Neural Networks, Long Short-Term Memory, and Bidirectional LSTM, CNN, and LSTM are used, and LSTM achieved 66 % accuracy.

The researchers in Hosseinmardi et al. (2015) studied cyberbullying on Instagram with 3000 images as a dataset and their associated comments. This study is based on previous work for cyberbullying, NLP, and sentiment analysis methodology. The study collected data and counted the number of bullying, not-bullying, and neutral comments.

In Arathi Unni et al. (2021), cyberbullying was studied based on the comments using NLP and ML. Different supervised ensemble learning techniques were used to classify comments. Support Vector Machine, Logistic Regression, and Perceptron models predicted the outputs based on the highest majority and achieved an accuracy of 94 %.

Using various state-of-the-art pre-trained models, the authors[19] worked on cyberbullying on images based on the attached text (captions and comments). Besides the deep learning approach, the authors used machine learning classifiers (Support Vector Machines and Logistic Regression).

The current literature on cyberbullying detection presents diverse approaches, focusing on different aspects of cyberbullying and employing various computational techniques. A summary of these studies is presented in Table 1.

The current literature on cyberbullying detection presents diverse approaches, focusing on different aspects of cyberbullying and employing various computational techniques. A summary of these studies is shown in Table 1. Several studies have explored the use of Natural Language Processing (NLP) and Machine Learning (ML) techniques for cyberbullying detection, with some focusing on text-based cyberbullying (Ali et al., 2018; Kansara & Shekokar, 2015) and others applying these techniques to images with associated text (Hosseinmardi et al., 2015; Zhong et al., 2016). Accuracy results, where reported, vary considerably, with a high of 95 % reported for caption-based detection (Zhong et al., 2016) and a low of 66 % using LSTM and CNN (Jadhav et al., 2023).

Several researchers have applied transfer learning techniques to the problem of image-based cyberbullying detection (Chakravarthi et al., 2021; Hitkul et al., 2019). These studies have shown promising results, with validation accuracy ranging from 65.62 % with InceptionV3 (Hitkul et al., 2019) to a high training and validation accuracy of 0.87 and 0.92, respectively, with ResNet50 (Chakravarthi et al., 2021). Studies have also examined the use of supervised machine learning models, such as SVM, Random Forest, and Logistic Regression, applied to text (Arathi Unni et al., 2021; Muneer & Fati, 2020; Talpur & O'Sullivan, 2020) and images (Kansara & Shekokar, 2015; Rosa et al., 2018). These studies reported high accuracy rates, up to 94 % (Arathi Unni et al., 2021).

Despite these advances, there remains a gap in the literature concerning the direct comparison of deep learning and traditional machine learning models for cyberbullying detection, mainly when applied to a multimodal classification approach combining text and images (Vishwamitra et al., 2021). Further, studies have yet to fully explore the potential of deep learning models as feature extractors for traditional machine learning classifiers in the context of cyberbullying detection. This research aims to fill this gap and contribute to the ongoing development of more effective cyberbullying detection methods.

The extensive literature review highlighted in Table 1 underlines the diverse methods currently employed for cyberbullying detection, including Natural Language Processing (NLP), Machine Learning (ML), and Transfer Learning techniques. While these studies have shown

Table 1
Literature review summary.

Paper	References	Type of Cyberbullying	Approach	Accuracy
A Framework for Cyberbullying Detection in Social Network	Kansara and Shekokar (2015)	Text and images	BOVW and (SVM) classifier	-
Content-Driven Detection of Cyberbullying on the Instagram Social Network	Zhong et al. (2016)	Images with text	Natural Language Processing, BoW, and Word2Vec classifying. By using Captions and DL-FS.	Overall, 68.55 %, using captions 95.00 %.
Cyberbullying Detection: An Overview	Ali et al. (2018)	text	Natural Language Processing and Machine Learning	-
Maybe look closer? Detecting trolling-prone images on Instagram	Hitkul et al. (2019)	images	Transfer Learning and Machine Learning	Inception V3: 65.62 val_acc, VGG16: 61.81 test acc
Cyberbullying severity detection: A machine learning approach	Talpur and O'Sullivan (2020)	text	Machine Learning classifiers – supervised learning – SVM & Random Forest	89 % to 91 %
A comparative analysis of machine learning techniques for cyberbullying detection on Twitter	Muneer and Fati (2020)	text	Machine learning – Logistic Regression	90.57 %
Meme classification using deep learning	Chakravarthi et al. (2021)	images	Transfer Learning – resnet50	(training: 0.87) and (validation 0.92)
Early detection of cyberbullying on social media networks	López-Vizcaíno et al. (2021)	text	Supervised Learning	-
Towards Understanding and Detecting Cyberbullying in Real-world Images	Vishwamitra et al. (2021)	Images based on text	Multimodal classification	93.36 %
CYBERBULLYING DETECTION	Jadhav et al. (2023)	text	LSTM, CNN, LSTM	66 %
Detection of Cyberbullying Incidents on Instagram Social	Hosseinmardi et al. (2015)	Images based on text-comments	NLP and sentiment analysis methodology	-
Detecting the Presence of Cyberbullying using Machine Learning	Arathi Unni et al. (2021)	text	SVM, Logistic Regression, and Perceptron models	94 %
A "Deeper" Look at Detecting Cyberbullying in Social Networks	Rosa et al. (2018)	Text and images	Deep learning and machine learning (Support Vector Machines and Logistic Regression)	-

promising results, a gap in the literature persists: few have examined the potential of deep learning models used as feature extractors for traditional machine learning classifiers in cyberbullying detection.

Therefore, this study proposes combining deep learning and traditional machine learning strengths. This hybrid method involves the application of deep learning models such as InceptionV3, ResNet50, and VGG16 as feature extractors, with the extracted features then serving as inputs for traditional machine learning classifiers, including Logistic Regression, Random Forest, and Support Vector Machines. By bridging the gap in the current literature, this research aims to contribute to the broader effort to enhance the effectiveness of cyberbullying detection methods.

3. Methodology

Our proposed approach for detecting cyberbullying in social media images combines deep learning and machine learning techniques in a synergistic framework. The overall workflow is illustrated in Fig. 2. First, input images undergo preprocessing steps like resizing and normalization. Next, we leverage the representation learning capabilities of deep convolutional neural networks (CNNs) using pre-trained models like VGG16 and ResNet50 as feature extractors. These models analyze the preprocessed images and output informative features capturing visual characteristics. The features from different CNNs are concatenated to create a joint representation, fusing knowledge from diverse sources. This combined feature set becomes the input to train traditional machine learning classifiers such as support vector machines and logistic regression. Finally, the trained model categorizes new images as containing cyberbullying or not based on learned patterns in the feature representations. This hybrid approach aims to leverage the complementary strengths of deep representation learning and efficient machine learning classification. In the following subsections, we delve into the details of each stage.

As aforementioned, earlier works on cyberbullying focused on the textual case or images with captions and comments. As a result, the analysis of the photos themselves is limited. The proposed methodology is initiated by making a dataset passed through pre-trained nets (e.g., VGG16, VGG19, Resnet50,.) and to see which learners among them will do the best. Secondly, the methodology is designed to use the pre-trained nets as a feature extraction tool to pass these features into machine learning classifiers (Random Forest, Logistic Regression, and Support Vector Machine) and see what accuracy will emerge, as shown in Fig. 3.

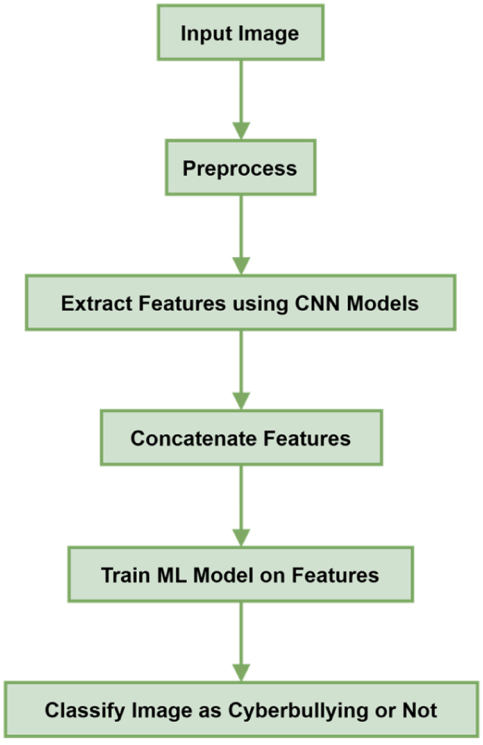


Fig. 2. Flowchart of proposed cyberbullying detection approach.

Fig. 2 shows the overall methodology for cyberbullying detection

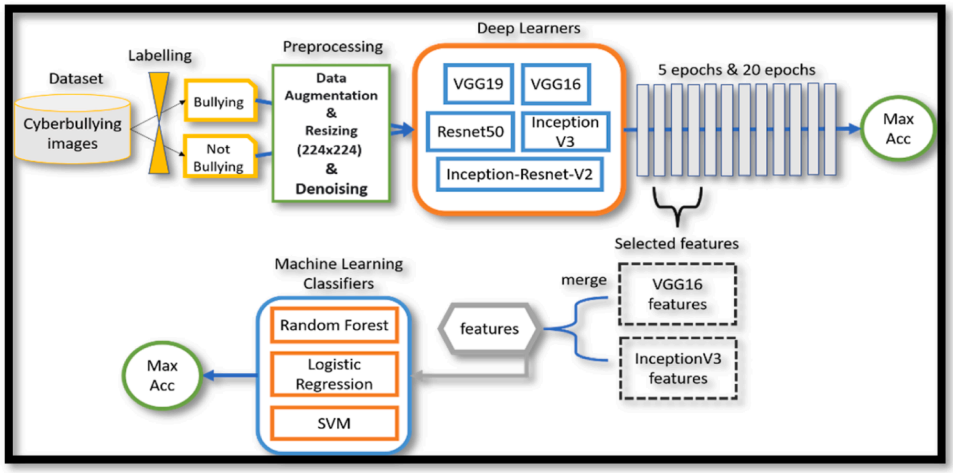


Fig. 3. Cyberbullying using images-methodology-our proposed.

using a hybrid deep learning and machine learning approach. It consists of the following key stages (Table 2):

1. Cyberbullying images: This is likely the dataset containing images that need to be classified into 'Bullying' or 'Not Bullying'.
2. Data Augmentation & Resizing (224 × 224) & Denoising: This step involves preprocessing the images to improve the model's performance. Data augmentation includes techniques like rotating, flipping, or zooming images to create a more robust dataset. Resizing standardizes image dimensions to 224 × 224 pixels, a common size for convolutional neural networks (CNNs). Denoising helps remove image noise to make patterns more distinguishable for the model.
3. VGG19, VGG16, Resnet50, Inception V3, Inception-Resnet-V2: These are different types of CNN architectures used for feature extraction. VGG19 and VGG16 are known for their simplicity and depth, Resnet50 uses residual connections to enable training of very deep networks, and Inception models apply filters of different sizes at the same level.
4. Features: The output from the CNN architectures. These are high-level features extracted from the images that represent the data in a form that's easier for the models to process and learn from.
5. Random Forest, Logistic Regression, SVM (Support Vector Machine): These are machine learning algorithms used for classification. Random Forest is an ensemble method using multiple decision trees, Logistic Regression is used for binary classification tasks, and SVM is effective in high-dimensional spaces.
6. Max Acc (Maximum Accuracy): This represents the goal of the workflow, which is to maximize the model's accuracy in classifying images correctly as either 'Bullying' or 'Not Bullying'. The term suggests that the best-performing model or ensemble of models will be chosen based on the highest accuracy metric.

Table 2
Summarizes the key acronyms and mathematical notations used throughout this paper.

Acronym/ Notation	Definition
CNN	Convolutional Neural Network
VGG19	19-layer Visual Geometry Group Network
VGG16	16-layer Visual Geometry Group Network
Resnet50	50-layer Residual Network
Inception V3	Version 3 of the Inception Network
Inception-Resnet-V2	Hybrid Network combining Inception and Residual Networks Version 2
SVM	Support Vector Machine
LR	Logistic Regression
RF	Random Forest
ACC	Accuracy
AUC	Area Under ROC Curve
ROC	Receiver Operating Characteristic

Cyberbullying detection requires understanding highly complex and nuanced contexts within images, a task difficult for machine learning or deep learning alone. A hybrid approach combines complementary strengths - machine learning provides interpretability and sample efficiency while deep learning enables learning robust representations straight from pixels. Fusing deep learning as a feature extractor and machine learning for classification harnesses their mutual benefits. However, social media images exhibit high variation, needing models that generalize well from limited data. Moreover, subtle gestures or sarcasm prove challenging, requiring contextual understanding beyond visual patterns. Deep learning models also tend to overfit with extensive training, reducing generalizability. The proposed scheme addresses these challenges through machine learning's ability to better generalize from small data and deep learning's representation learning, converting raw pixels into high-level concepts. Together, the hybrid approach is better equipped to comprehend complex contexts like sarcasm. The machine learning component also acts as a regularizing influence, preventing deep learning overfitting. In summary, the hybrid technique is motivated by the complementarity of the machine and deep learning and overcomes key challenges in cyberbullying detection, including generalization, contextual understanding, and overfitting.

3.1. *Preprocessing process*

Dataset: our work is based on a dataset from Instagram and Vine (Hosseinmardi et al., 2015). The research team labeled their data using majority voting, which means that they showed the same image to five different people and asked whether it represented cyberbullying after that taking the majority of their voting, using the CrowdFlower, a crowd-sourced website for cyber-aggression and cyberbullying. The data combines the extracted images and their captions with the associated comments. The researchers have concluded that the number of words increases with the decrease of the media sessions. The resulting dataset has shrunk to 2300 images.

Dataset Labelling: There are still points of difference and controversy in classifying actions in common as bullying. This difference might be attributed to culture, customs, norms, and traditions. It sometimes might be related to other matters such as religion and belief, as what some person sees as bullying may not be seen by another person. From this point of view, the present study had to define the criteria for the individuals who participated in classifying the images as bullying or not

before they started the mission. They underwent prior (training) sessions to identify bullying in pictures, and many discussions took place at this stage for everyone to have a clear idea about the true meaning of bullying and the featured points that make the image attract bullying or not. For example, images related to "drugs" are classified as "cyberbullying", whereas images that are related to "tattoos" and "food" are not classified as cyberbullying. Most of the bullying rate came as aggressive notes against women; An image of a woman attracts a huge number of cyberbullying comments while that image is just a normal selfie image.

Fig. 4 shows some examples of cyberbullying labelling processes accomplished in this paper. As in samples (a) and (b), mockery is labelled as bullying, which often uses changing something of the image or drawing on it as some sign of bullying. Whereas (h) is a picture of someone practising sports labelled as not bullying. Drugs and smoking threats always represent bullying; normal selfies and sports or food are not bullying. But, a picture of a woman sometimes attracts bullying on it.

Dataset Augmentation and preprocessing: Three main processes are used:

Rotation: for all images in the dataset, we applied three rotation angles (90, 180, -90), using the method rotate (θ) given in Eq. (1).

$$\text{Rotation} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{1}$$

Flipping: flipping the images horizontally and vertically based on the formulation in Eq. (2).

$$\text{Flipping} = \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \tag{2}$$

Resizing: resizing all images to 224 × 224 to fit the pre-trained nets.

Denoising: some images have a poor resolution or distortion; denoising can filter and improve the image.

As a result of the augmentation process, the dataset contains 13554 images from the augmented 2300 images.

Classification of Bullying in Images: This type of classification might be easy for human beings, but for a computer program, it would be too difficult because even if we can create a smart algorithm to detect some special features of the image (e.g., the posture of hands), it will remain challenging as this part may be found larger or smaller than the one programmed. Even though there are still too many possibilities in an image, such as direction or flipping, which will increase the algorithm's complexity to do the detection, Convolutional Neural Network (CNN) brought the solution; we do not have to explicitly tell the computer the difference between the images that have Bullying or Not-bullying. Instead, you just feed enough images of both Bullying or Not-bullying into the CNN. It will automatically do the magic and generate a correct neural network structure to match the problem.

There is a way of using deep learners as feature extractors when getting the features from more than one deep learner and merging them into one array of features. Give them as inputs to machine learning classifiers and train the model using them.

3.2. Deep learning and artificial neural network

Deep learning algorithms show superior learning and classification performance in many areas, like transfer learning and recognition in many types, such as speech and objects (Engineers, 2012). It has been approved that most classification problems can be solved using deep learning more than machine learning classifiers. Deep learning uses the Artificial Neural Network (ANN), a successful model inspired by the human brain and its functionalities (Pang et al., 2020). ANN was proposed to solve various complex problems in the last century, but now they are used for almost everything. ANN consists of the input layer, output layers, and hidden layer. Inputs are multiplied by weights and entered into the next layer as inputs. Sometimes, hidden layers contain activation functions according to the classification problem. In the end, the sum of all resulted values from every branch from the input layer to the output layer will be the output of the ANN as shown in Fig. 5.

Deep learning uses ANN with more hidden layers and various activation functions, but it gives more accuracy when dealing with large-scale datasets. It has an easy and simple implementation of ANN and deep learning approaches. Also, it has supported the Convolutional Neural Networks and made it easier to do the transfer learning to make the analysis faster.

3.3. Transfer learning

A circumstance in which what one has learned in one setting is used in another is referred to as "transfer learning" and "domain adaptation." (Goodfellow et al., 2016). Transfer learning is a machine learning



Fig. 4. Cyberbullying labeling.

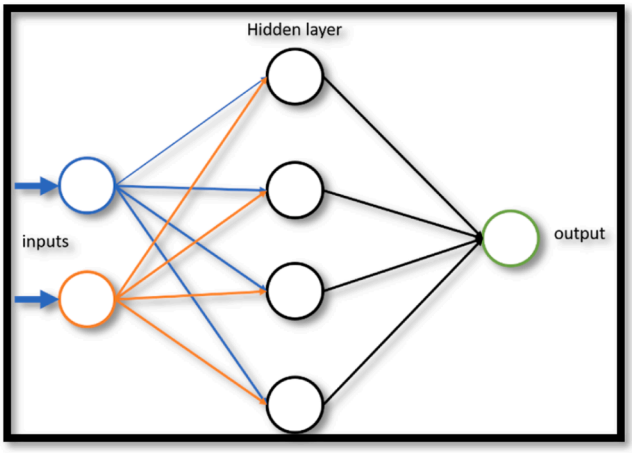


Fig. 5. Artificial neural network architecture.

technique in which a model generated for one job is utilized as a starting point for a model developed for another activity. So, instead of starting from scratch, it utilizes a pre-trained model to alter the final layer and feed it with data.

An image identification technique known as training deep learning models was adapted for this study by tweaking a few of its parameters. In this work, a trained learner, such as VGG16, is employed and retrained to adapt to the situation. This process is known as transfer learning.

Convolutional Neural Network (CNN): CNN is a deep learning model inspired by the structure of the animal visual cortex for processing data with a grid pattern, such as photographs. A CNN aims to learn spatial feature hierarchies automatically and adaptively through back-propagation by using multiple building blocks, such as convolution layers, pooling layers, and fully connected layers (Yamashita et al., 2018). See Fig. 6, which shows the connected layers of CNN.

In a CNN, there are three layers: convolutional, pooling, and fully connected, as seen in Fig. 5. The backpropagation technique trains a set of learnable weights that form the filters in a convolutional layer. Filters are like repositories, and when they are concatenated with the input tensor, they try to detect patterns in the input that are similar to a stored pattern. It is possible to use the filtering system without manually choosing the features you want to use. The feature maps are the output of one filter applied to the preceding layer, which comprises the second part of the convolutional layer. When a specific filter is drawn across the preceding layer, it activates the neuron in that location, and the data is collected in an output map. When activated, the stride is the distance a filter moves across the input from the preceding layer. Mock inputs can be created with zero values using techniques like zero-padding if the previous layer is too large to divide evenly by the filters and stride sizes. If so, the input feature map can be read off its edge.

The second layer that CNN includes is the pooling layer, which performs subsampling, down-sampling, and dimension reduction. To consolidate the features learned and expressed in the preceding layer's feature map, pooling layers sample the previous layer's feature map and apply one or more convolutional layers. Different types of pooling exist, such as maximum, average, and sum.

The third type is the fully connected layers used for mapping and classification. The output is flattened and fed into a fully connected layer (FC Layer). At the feedforward neural network layer, FC layers are the norm. Class prediction probabilities can be generated using either a nonlinear activation function or a Softmax activation in these layers. Convolution and pooling layers are used to collect and consolidate data before FC layers are used to construct final nonlinear combinations of features and make predictions by the network. FC layers are utilized after the network.

There are other pre-trained models in ImageNet, and they are the most popular VGG: Fig. 6 shows that increasing the weight layer depth to 16–19 shows a significant improvement over prior-art configurations when using an architecture with very small (33) convolution filters, a technique known as the popular VGG. The ImageNet Challenge 2014 submission was based on these findings. Deep visual representations in computer vision have been made publicly available by the VGG team in the form of two of their best-performing ConvNet models (Simonyan and Zisserman, 2014). Figs. 7 & 8 depicts the design of VGG16 in its entirety.

Resnet50: This is a very deep CNN design that uses residual learning to try to overcome the degradation issue. The output y has an additional input x so that a function H(x) is redefined as:

$$H(x) = F(x) + x, \tag{3}$$

So that makes the learning process easier. From its name, it has 50 CNN-connected layers, and in general, it passed through good training and validation accuracy.

Table 3 provides detailed information about various deep-learning models. Specifically, it lists and compares the VGG16, VGG19, ResNet50, InceptionV3, and InceptionResNetV2 models across several different specifications, including the size of the model (in megabytes), Top-1 and Top-5 accuracy, the number of parameters, the depth of the model, and the time it takes to perform an inference step on both CPU and GPU platforms.

From the table, VGG16 and VGG19 are the largest models in size and

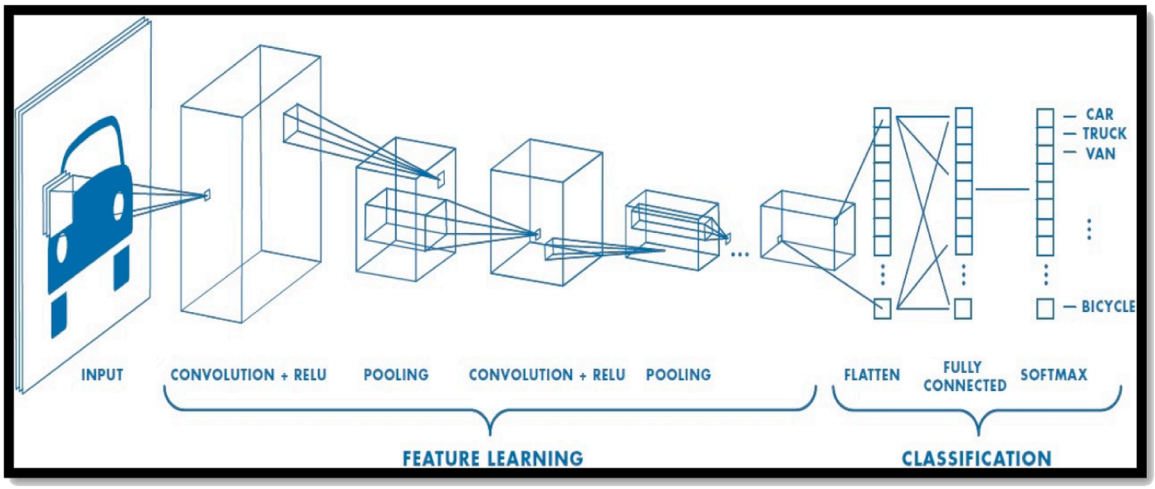


Fig. 6. CNN Diagram.

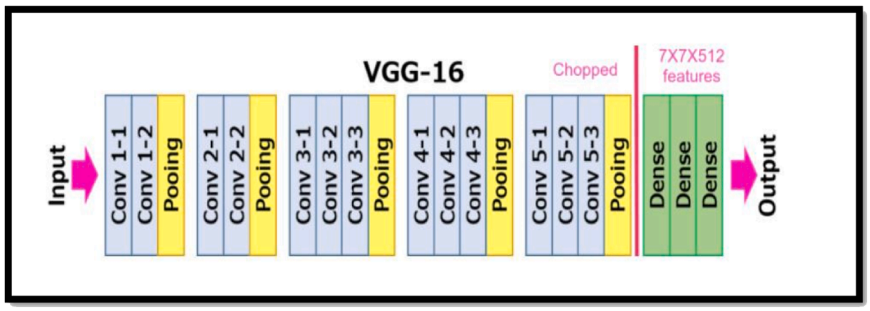


Fig. 7. VGG16 layers (Prabhu 2018; LaptrinhX 2020).

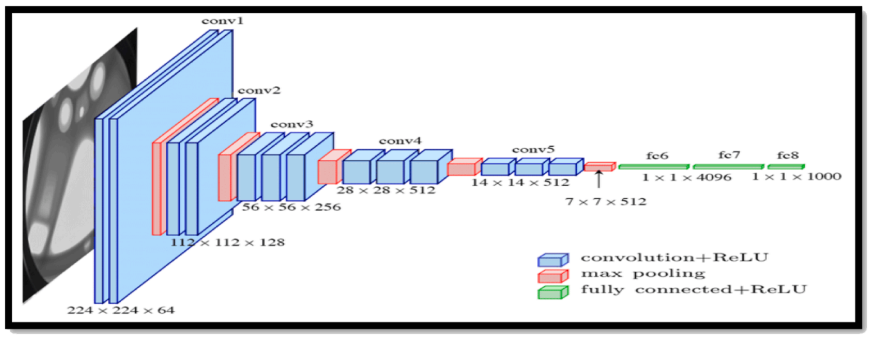


Fig. 8. VGG16 Architecture (Ferguson et al., 2017).

Table 3
Deep Learners specifications.

Model	Size (MB)	Top-1 Accuracy	Top-5 Accuracy	Parameters	Depth	Time (ms) per inference step (CPU)	Time (ms) per inference step (GPU)
VGG16	528	0.713	0.901	138,357,544	23	69.50	4.16
VGG19	549	0.713	0.900	143,667,240	26	84.75	4.38
ResNet50	98	0.749	0.921	25,636,712	-	58.20	4.55
InceptionV3	92	0.779	0.937	23,851,784	159	42.25	6.86
InceptionResNetV2	215	0.803	0.953	55,873,736	572	130.19	10.02

parameter count, yet they do not provide the highest Top-1 and Top-5 accuracies. They are also the slowest in CPU inference time but relatively fast on the GPU. ResNet50, while significantly smaller in size and with fewer parameters than the VGG models, achieves better Top-1 and Top-5 accuracies. It also performs inference faster on the CPU but slightly slower on the GPU. InceptionV3, the most miniature model with minor parameters, achieves better accuracy than VGG16, VGG19, and even ResNet50. It shows the best performance in CPU inference time but is slower on the GPU. InceptionResNetV2, although more significant in size and parameter count than ResNet50 and InceptionV3, provides the highest Top-1 and Top-5 accuracies among all the models. However, it also requires the longest inference time on both CPU and GPU.

3.4. Machine learning classifiers

This study will use supervised learning from machine learning since our datasets are labelled into two classes. Three algorithms are selected from the highest-performing classifiers for categorical classification.

Random Forest (RF): RF (Belavagi and Muniyal, 2016) is a classification tree algorithm. Forest-based classifiers will benefit significantly from this algorithm. During the classification process, there is no re-modification of the dataset. Hence, it can be used to manage dataset noise values with an acceptable accuracy rate. When implementing this algorithm, ensure to account for the number of trees in the forest, as each tree in the forest forecasts the predicted outcome. Afterwards, a voting method is utilized to pick the expected result with the most votes.

Logistic Regression (LR.): It is a classifier more than a predictor; the LR model classifies data into two classes: yes/no, and 0/1. It is a good classifier that sometimes yields a good performance (Nusinovici et al., 2020); it usually uses the sigmoid function to predict the class:

$$\log\left(\frac{p}{1-p}\right) \tag{4}$$

The data behaviour is represented in Fig. 9.

Support Vector Machines are a popular machine learning algorithm commonly used for classification tasks. The key idea behind SVMs is to

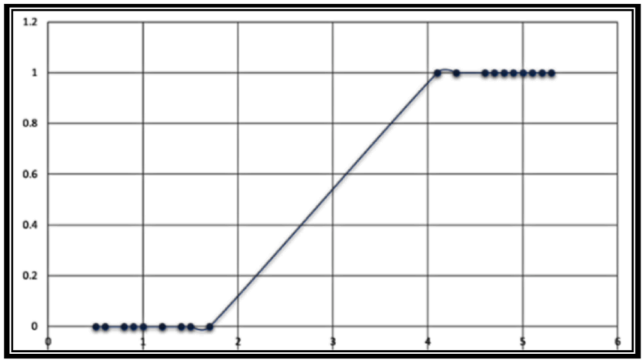


Fig. 9. Logistic regression.

find a hyperplane or boundary that separates the data points into different classes. This is achieved by maximizing the margin between the hyperplane and the nearest data points on each side. These closest data points are called support vectors and dictate where the hyperplane is positioned. SVMs can handle nonlinear separations by using kernel functions to map the data into higher dimensional spaces where separation may be easier. Overall, SVMs aim to create the optimal separating hyperplane that maximizes the distance to the support vectors, minimizing generalization error. This makes them effective for learning from limited training data. Cervantes et al. (2020) See Fig. 10: SVM-Maximum margin between classes.

4. Experiments and results

Environments like PyCharm and Anaconda are appropriate for implementing the model. However, using cloud computing achieves high performance in analyzing and modelling. Google Colab provides all requirements and reduces the time needed to install the required libraries and models.

In the first experiment, labelled images were fed into five pre-trained models: VGG16, VGG19, ResNet50, InceptionV3, and InceptionResNetV2. The SoftMax activation function was utilized, given its superior performance over ReLU and sigmoid functions. The batch size was 32, and the models were trained over 5 epochs. The results of this experiment are summarized in Table 4.

The second experiment started by performing the same operation as in the first experiment, but 20 epochs were used instead of 5. As known, one of the critical issues when training a neural network model is overfitting (Devarakonda et al., 2017). Increasing the number of epochs means that the model will be specific for a distinct sample of data, which makes the model incapable of another dataset. This model achieved high training accuracy but failed on the test part (Agrawal et al., 2022; Khanduzi & Sangaiah, 2023; Sajeeda & Hossain, 2022; Sangaiah et al., 2022, 2023). Therefore, high accuracy does not always mean the model is doing good. Table 5 Deep learners (20 epochs) show the result of this experiment.

The data suggests a trade-off between model size, parameter count, accuracy, and inference time. Larger models, such as VGG16 and VGG19, do not necessarily yield better accuracy. Compact models like InceptionV3 can achieve higher accuracy with fewer computational resources. However, the model that delivers the highest accuracy, InceptionResNetV2, requires the most resources and time.

As shown in Fig. 11, The bar charts compare the performance of various deep learning networks, specifically focusing on Accuracy and Loss metrics, after training for 5 and 20 epochs. The blue bars represent results from 5 epochs, while the green bars show the outcomes after 20 epochs, highlighting the impact of extended training on model performance.

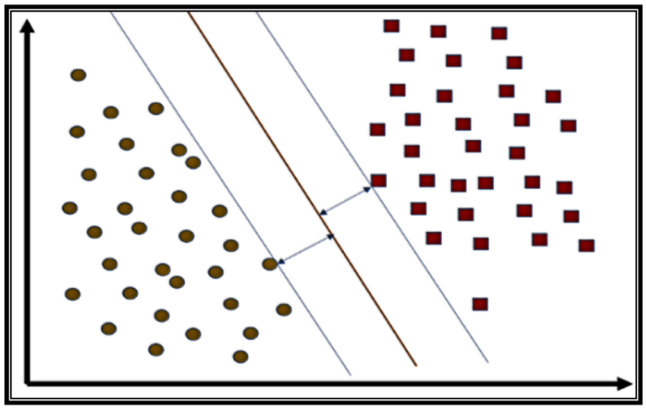


Fig. 10. The SVM-Maximum margin between classes.

Table 4
Deep learners (5 epochs).

Net (5 epochs)	Accuracy	Loss	Val_accuracy	Val_loss
VGG16	0.8451	0.3638	0.5826	0.9358
InceptionResnetV2	0.7882	0.8864	0.6348	2.3698
VGG19	0.7980	0.4700	0.5739	0.8898
Resnet50	0.7108	0.6933	0.6348	0.7591
InceptionV3	0.7873	1.1490	0.6522	3.6272

Table 5
Deep learners (20 epochs).

Net (20 epochs)	Accuracy	Loss	Val_accuracy	Val_loss
VGG16	0.9450	0.1567	0.6261	1.2193
InceptionResnetV2	0.8801	0.5370	0.6261	2.5083
VGG19	0.9146	0.2037	0.6174	1.1616
Resnet50	0.6945	0.9416	0.6348	1.2697
InceptionV3	0.9030	0.8024	0.6087	6.8953

As shown in the two tables above, the inceptionV3 has achieved the highest accuracy (65 %) in the 5 epochs, whereas the Resnet50 has the highest performance among the selected models with 63 % in the 20 epochs. Fig. 12: 5 & 20 epochs accuracy shows the difference between the 5 and 20 epochs accuracy.

Fig. 13, Train accuracy, train loss, validation accuracy, and validation loss show the performance of each value of training accuracy, training loss, validation accuracy, and validation loss on their growth with the training process for the best three deep learners (VGG16, InceptionV3, and Resnet50).

As shown in Fig. 13, the training accuracy was more significant than the validation accuracy. However, the validation loss was more effective than the training loss. A deep learning approach is not enough to detect cyberbullying accurately.

The third experiment used the models with the highest accuracy (InceptionV3, resnet50, and VGG16) as feature extractors by getting their features and merging them into one array. Then, feed them as inputs for the classifiers trained using machine learning (Logistic Regression, Random Forest, and Support Vector Machine). Table 6: Machine Learning classifiers results show the results for applying the features extracted from VGG16 and feeding the ML classifiers.

Fig. 14: confusion matrix (LR, RF & SVM) represents the three implemented classifiers in terms of the confusion matrix.

The experiments have shown that machine learning classifiers such as logistic regression achieved 67 % accuracy, whereas deep learning has reached 65 % from InceptionV3 deep learner. Moreover, machine learning has a higher performance than deep learning in this problem of detecting cyberbullying from images. In addition, the testing part showed promising results with an overall accuracy of 67 %. Fig. 15 shows samples of cyberbullying detection system testing.

We conducted extensive experiments to validate our approach under varied settings. Key parameters and their values are shown in Table 7. We used a batch size of 32 based on GPU memory constraints, and a learning rate of 0.001 which provided optimal convergence. The CNN architectures used are specified in Table 1. We strategically regularized training to prevent overfitting, stopping at 50 epochs.

5. Discussion

The key findings from the experiments show a trade-off between model size, parameter count, accuracy, and inference time. Larger models, such as VGG16 and VGG19, do not necessarily yield better accuracy. Compact models like InceptionV3 can achieve higher accuracy with fewer computational resources. However, the model that delivers the highest accuracy, InceptionResNetV2, requires the most resources and time. From the results obtained in the 5-epoch run (Table 3),

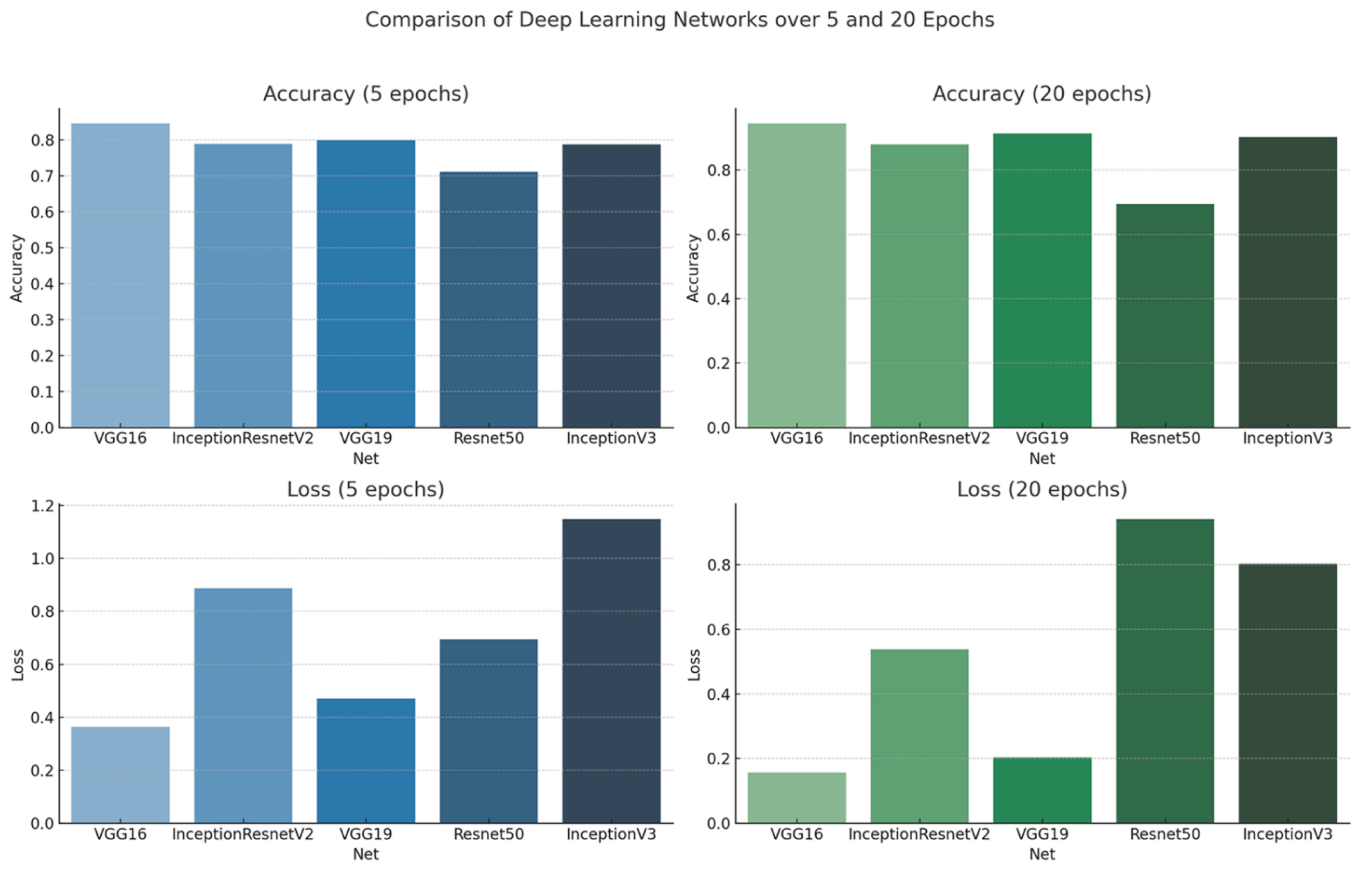


Fig. 11. Compares the performance of various deep learning networks.

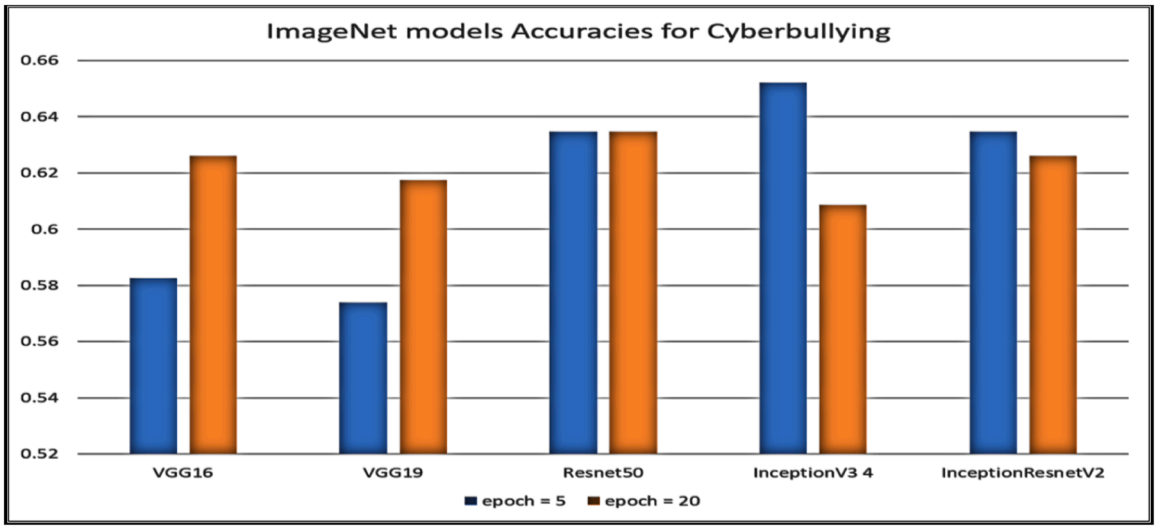


Fig. 12. 5 & 20 epochs accuracy.

InceptionV3 outperformed the others with a validation accuracy of 65 %. On the other hand, for the 20-epoch run (Table 4), ResNet50 demonstrated the highest validation accuracy among the models at 63 %.

The results also illustrate the overfitting problem in deep learning. As the number of epochs increased, the model's performance on the training set improved, but its performance on the test set declined. This suggests that a balance must be struck when choosing the number of epochs to avoid overfitting. Finally, the third experiment demonstrated the potential utility of deep learning models as feature extractors for traditional machine learning classifiers. Logistic Regression and Support Vector Machine classifiers achieved an accuracy of 67 % when trained on the features extracted by the deep learning models.

This study's main challenge was managing the trade-off between model complexity, computational resources, and performance. High-performing models require substantial computational resources, which may not be feasible for real-time applications. Moreover, the study demonstrated the common issue of overfitting in deep learning. This indicates the need for regularization, early stopping, or more sophisticated validation techniques to improve the model's generalization

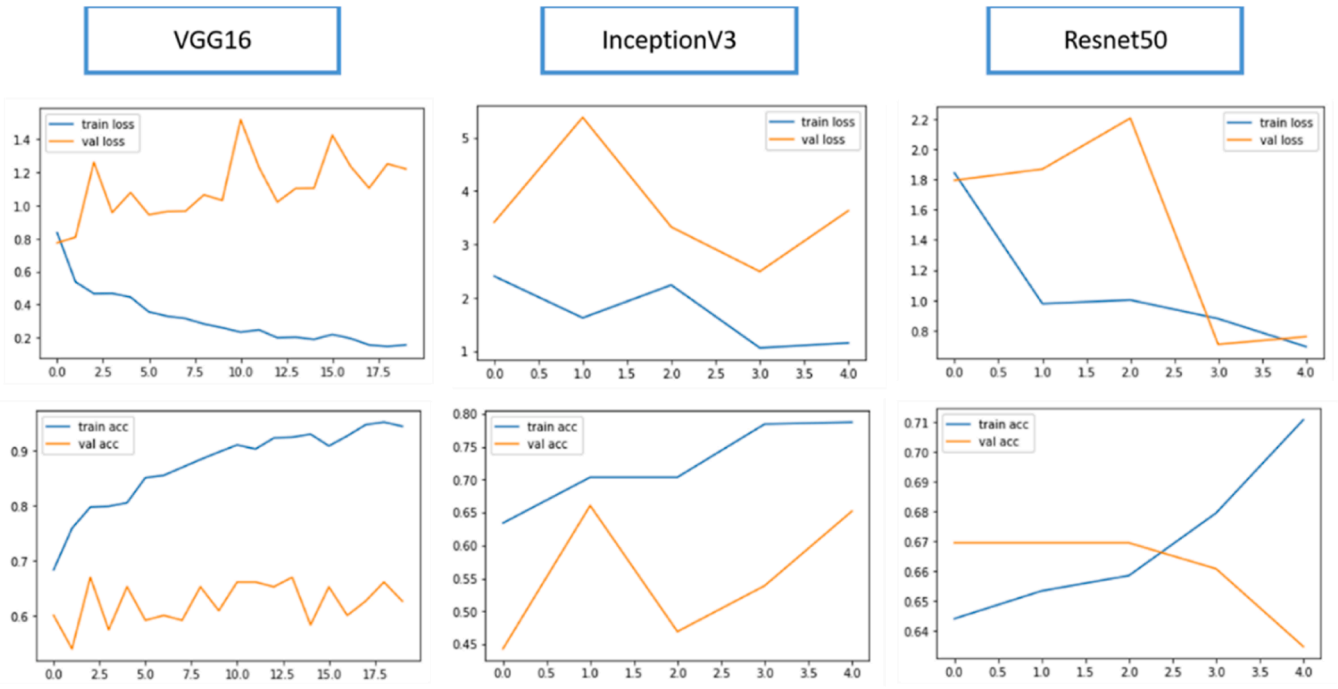


Fig. 13. Train accuracy, train loss, validation accuracy, and validation loss.

Table 6
Machine Learning classifier's results.

ML Classifier	Logistic regression	Random forest	Support vector machine
Accuracy	0.669	0.617	0.669

ability. Lastly, while the deep learning models showed promise as feature extractors for machine learning classifiers, the performance was still not optimal. Future work could explore different architectures or fine-tuning strategies to improve performance.

The key findings of this research can be summarized as follows: InceptionV3 outperformed others in the 5-epoch experiment with a validation accuracy of 65 %. However, for the 20-epoch experiment, ResNet50 emerged as the top performer with a 63 % validation accuracy, indicating different models' effectiveness depends on the number of epochs. Overfitting was a significant issue in the 20-epoch experiment, underscoring the models' tendency to lose performance on the test set with increased training. Additionally, employing deep learning models as feature extractors for traditional machine learning classifiers, such as Logistic Regression and Support Vector Machines, achieved a promising 67 % accuracy, suggesting a potential advantage in blending deep learning with traditional machine learning techniques in specific applications.

This hybrid framework demonstrates the promising potential for multi-platform cyberbullying detection. The model can be extended to other social media sites such as Twitter, Facebook, and Reddit by retraining on relevant image data from those platforms. Furthermore, enhancing the model to process text, audio, and video data would allow holistic detection across diverse online content. Technically, employing regularization strategies and validating performance on varied datasets will improve generalization.

Exciting real-world applications include integrating such systems into social media and gaming platforms to provide live moderation and promote safety. Tailoring the model for different demographics and cultures through personalized training is another impactful direction. Beyond cyberbullying, the hybrid approach can be applied to a range of problematic online behaviours like hate speech, misogyny, and misinformation. Advancing multidisciplinary research combining ethics, education, and technology will enable the responsible and transparent development of such AI systems.

Releasing the model architecture, training methodology, and datasets to the research community will accelerate progress in this critical

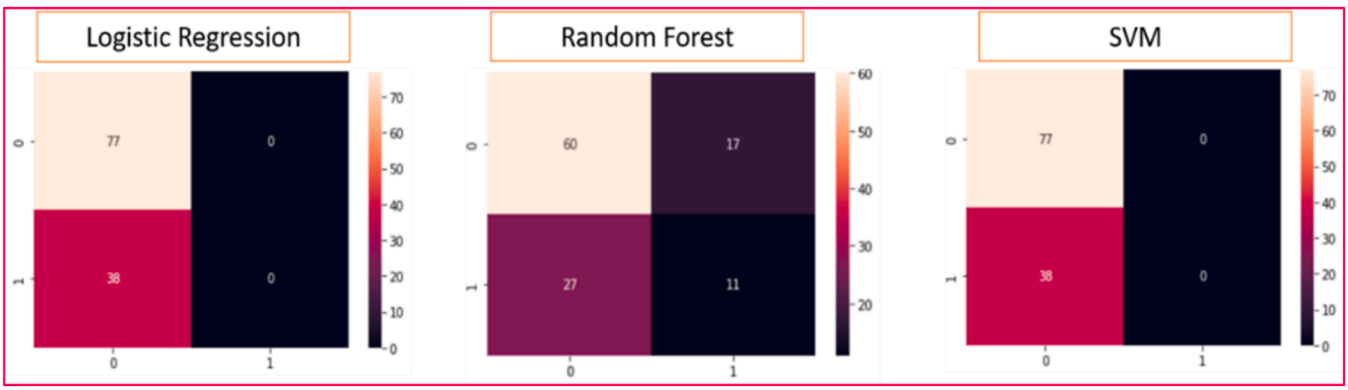


Fig. 14. Confusion matrix (LR, RF & SVM).

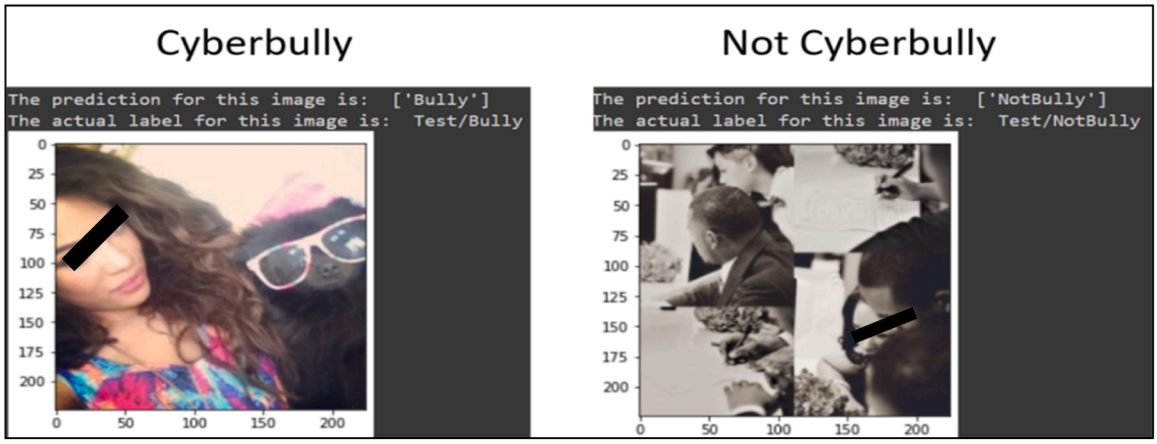


Fig. 15. Samples of cyberbullying detection system testing.

Table 7
Model parameters.

Parameter	Value
Batch Size	32
Learning Rate	0.001
CNN Architectures	VGG16, ResNet50, InceptionV3

area. With thoughtful and inclusive design, hybrid deep learning and machine learning techniques can help address online harms. But fundamentally, cultivating empathy and compassion in our societies is critical to resolving such issues in the long term.

While providing improved detection accuracy, a key challenge of our framework is the computational overhead incurred. Employing multiple pre-trained deep neural networks for feature extraction adds substantial processing and memory requirements during training and inference due to their millions of parameters. Furthermore, fusing features from different models expands the dimensionality, increasing the load when training machine learning classifiers. However, recent advances in parallel computing, model optimization, and cloud infrastructure help mitigate these demands. We aim to conduct efficiency analyses in future work to quantify our approach's resource utilization and runtimes. Optimizing model architectures and exploring knowledge distillation could potentially reduce overhead. Overall, some increase in complexity is justified, given the significant performance gains achieved. However, developing efficient implementations remains essential for real-world viability.

In summary, while the hybrid technique does increase computational costs, this overhead is warranted, given the accuracy benefits demonstrated. We acknowledge efficiency as an essential area for improvement. With optimized implementations and infrastructure, the higher resource requirements would be manageable. Please let me know if I should expand on any part of this added discussion about the complex trade-offs involved.

6. Conclusion

The popularity of social media and its improvement has increased immensely. However, with this immense growth and use, issues such as cyberbullying have risen. The problem severely impacts teens, children, and their parents and can lead to suicide. Therefore, the early detection of such behaviour can prevent cyberbullying or mitigate its effects. Computer vision has been used through a deep learning approach to achieve this goal. Using pre-trained and state-of-the-art models has made the modelling and analysis faster and more accurate.

Moreover, using these systems as feature extractors and passing the features to machine learning classifiers is another contribution of this work. The experiments have shown that the proposed approach using Random Forest, SVM, and Logistic regression classifiers achieved 67 % accuracy. In addition, the proposed method using deep learning with inceptionV3 has gained 65 % with five epochs. Machine learning performs better than deep learning in cyberbullying detection of images.

Despite the challenges, the potential of deep learning in applications like cyberbullying detection is clear. With continued advancements in model architecture, training strategies, and computational resources, we can expect to see even more impressive performance in the future. We plan to collect and label a new dataset based on each country's traditions and norms in future work. In addition, we plan to use recognition and object segmentation to recognize the posture of hands as an example of the bad signs that can be classified as cyberbullying. They can enhance the approach and remove them from social media, which may prevent this phenomenon from spreading.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Agrawal, P., Katal, N., & Hooda, N. (2022). Segmentation and classification of brain tumor using 3D -UNet deep neural networks. *International Journal of Cognitive Computing in Engineering, 3*, 199–210.

Aldweesh, A., Alauthman, M., Al Khaldy, M., Ishtaiwi, A., al-Qerem, A., et al. (2023). The meta-fusion: A cloud-integrated study on blockchain technology enabling secure and efficient virtual worlds. *International Journal of Cloud Applications and Computing (IJCAC), 13*(1), 1–24.

Ali, W. N. H. W., Mohd, M., & Fauzi, F. (2018). Cyberbullying detection: an overview. In *2018 Cyber Resilience Conference (CRC)* (pp. 1–3). IEEE.

Arathi Unni, R. K. R., Sebastian, Linda, Rajalakshmi, S, & Siby, Sissy (2021). Detecting the Presence of Cyberbullying using Machine Learning. *International Journal of Engineering Research & Technology (IJERT), 9*(13).

Belavagi, M. C., & Muniyal, B. (2016). Performance evaluation of supervised machine learning algorithms for intrusion detection. *Procedia Computer Science, 89*, 117–123.

(2022). Bullying Statistics-Cyber Bullying Statistics.

Cervantes, J., Garcia-Lamont, F., Rodríguez-Mazahua, L., & Lopez, A. (2020). A comprehensive survey on support vector machine classification: Applications, challenges and trends. *Neurocomputing, 408*, 189–215.

Chakravarthi, B. R., Priyadharshini, R., Krishnamurthy, P., & Sherly, E. (2021). In *Proceedings of the First Workshop on Speech and Language Technologies for Dravidian Languages*.

Constine, J. (2018). *Instagram now uses machine learning to detect bullying within photos.* Tech Crunch.

Dean, B. (2021). *Social network usage & growth statistics: How many people use social media in 2022*.

A. Devarakonda, M. Naumov, and M. Garland, "Adabatch: Adaptive batch sizes for training deep neural networks," arXiv preprint arXiv:1712.02029, 2017.

Engineers, I.o. E.a. E. (2012). Signal processing and their applications. In *2012 11th international conference on information science, signal processing and their applications (ISSPA)*. Piscataway, NJ: IEEE.

Ferguson, M., Ak, R., Lee, Y.-T. T., & Law, K. H. (2017). Automatic localization of casting defects with convolutional neural networks. In *2017 IEEE international conference on big data (big data)* (pp. 1726–1735). IEEE.

G. O. Broadband. (2022). *All the latest cyber bulling statistics and what they mean in 2022*. Available: https://www.broadbandsearch.net/blog/cyber-bullying-statistics.

Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. MIT press.

Hitkul, H., Shah, R. R., Kumaraguru, P., & Satoh, S.i. (2019). Maybe look closer? Detecting trolling prone images on Instagram. In *2019 IEEE fifth international conference on multimedia big data (BigMM)* (pp. 448–456). IEEE.

H. Hosseinmardi, S. A. Mattson, R. I. Rafiq, R. Han, Q. Lv, and S. Mishra, "Detection of cyberbullying incidents on the instagram social network," arXiv preprint arXiv: 1503.03909, 2015.

Jadhav, R., Chaudhari, G., & Rane, S. (2023). Cyber bullying detection. *International Journal of Creative Research Thoughts (IJCRT)*, 1–6.

Joshi, S., Koparrati, S., & Singhal, V. (2022). Marketing analytics for analyzing social media and branding efforts of Indian telecom operators. *International Journal of Cloud Applications and Computing, 12*(1), 1–18.

Kansara, K. B., & Shekokar, N. M. (2015). A framework for cyberbullying detection in social network. *International Journal of Current Engineering and Technology, 5*(1), 494–498.

Khanduzi, R., & Sangaiah, A. K. (2023). An efficient recurrent neural network for defensive Stackelberg game. *Journal of Computational Science*. , Article 101970. https://doi.org/10.1016/j.jocs.2023.101970

LaptrinhX. (2020). *Face recognition using transfer learning and VGG16*. Available: https://laptrinhx.com/face-recognition-using-transfer-learning-and-vgg16-3833175636/.

Li, H. (2015). *Image analysis of cyberbullying using machine learning techniques*.

Li, S., Qin, D., Wu, X., Li, J., Li, B., & Han, W. (2022). False alert detection based on deep learning and machine learning. *International Journal on Semantic Web and Information Systems, 18*(1), 1–21.

López-Vizcaíno, M. F., Nóvoa, F. J., Carneiro, V., & Cacheda, F. (2021). Early detection of cyberbullying on social media networks. *Future Generation Computer Systems, 118*, 219–229.

Muneer, A., & Fati, S. M. (2020). A comparative analysis of machine learning techniques for cyberbullying detection on Twitter. *Future Internet, 12*(11), 187.

Mussap, A. J., Clancy, E. M., & Klettke, B. (2023). Attitudes and beliefs associated with cyberbullying and non-consensual sexting in cisgender and transgender adults. *Gender Issues, 40*(1), 65–85.

Nusinovici, S., et al. (2020). Logistic regression was as good as machine learning for predicting major chronic diseases. *Journal of Clinical Epidemiology, 122*, 56–69.

Pang, Z., Niu, F., & O'Neill, Z. (2020). Solar radiation prediction using recurrent neural network and artificial neural network: A case study with comparisons. *Renewable Energy, 156*, 279–289.

Prabhu. (2018). *Understanding of Convolutional Neural Network (CNN) — Deep learning*. Available: https://medium.com/@RaghavPrabhu/understanding-of-convolutional-neural-network-cnn-deep-learning-99760835f148.

Rosa, H., Matos, D., Ribeiro, R., Coheur, L., & Carvalho, J. P. (2018). A "deeper" look at detecting cyberbullying in social networks. In *2018 International joint conference on neural networks (IJCNN)* (pp. 1–8). IEEE.

Sahoo, S. R., & Gupta, B. B. (2019). Hybrid approach for detection of malicious profiles in twitter. *Computers & Electrical Engineering, 76*, 65–81.

Sajeeda, A., & Hossain, B. M. (2022). Exploring generative adversarial networks and adversarial training. *International Journal of Cognitive Computing in Engineering, 3*, 78–89.

Sangaiah, A. K., Javadpour, A., Pinto, P., Ja'fari, F., & Zhang, W. (2022). Improving quality of service in 5G resilient communication with the cellular structure of smartphones. *ACM Transactions on Sensor Networks (TOSN), 18*(3), 1–23.

Sangaiah, K., Javadpour, A., Ja'fari, F., Zhang, W., & Khaniabadi, S. M. (2023). Hierarchical clustering based on Dendrogram in sustainable transportation systems. In *IEEE transactions on intelligent transportation systems*. https://doi.org/10.1109/TITS.2022.3222789

K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," arXiv preprint arXiv:1409.1556, 2014.

Sultan, T., Jahan, N., Basak, R., Jony, M. S. A., & Nabil, R. H. (2023). Machine learning in cyberbullying detection from social-media image or screenshot with optical character recognition. *International Journal of Intelligent Systems and Applications, 15*, 1–13.

Sun, N., Li, J., Liu, Y., & Arya, V. (2023). A scalable sharding protocol based on cross-shard dynamic transaction confirmation for alliance chain in intelligent systems. *International Journal on Semantic Web and Information Systems, 19*(1), 1–30.

Talpur, B. A., & O'Sullivan, D. (2020). Cyberbullying severity detection: A machine learning approach. *PLoS ONE, 15*(10), Article e0240924.

Upadhyay, U., Kumar, A., Sharma, G., et al. (2023). Cyberbullying in the metaverse: A prescriptive perception on global information systems for user protection. *Journal of Global Information Management, 31*(1), 1–25.

Vishwamitra, N., Hu, H., Luo, F., & Cheng, L. (2021). Towards understanding and detecting cyberbullying in real-world images. In *2020 19th IEEE international conference on machine learning and applications (ICMLA)*.

Yamashita, R., Nishio, M., Do, R., & Togashi, K. (2018). Convolutional neural networks: An overview and application in radiology. *Insights Imaging, 9*(4), 611–629.

Yi, P., & Zubiaga, A. (2023). Session-based cyberbullying detection in social media: A survey. *Online Social Networks and Media, 36*, Article 100250.

Zhang, T., Zhang, Z., Zhao, K., Gupta, B. B., & Arya, V. (2023). A lightweight cross-domain authentication protocol for trusted access to industrial internet. *International Journal on Semantic Web and Information Systems, 19*(1), 1–25.

Zhang, Z., Sun, R., Zhao, C., Wang, J., Chang, C. K., & Gupta, B. B. (2017). CyVOD: a novel trinity multimedia social network scheme. *Multimedia Tools and Applications, 76*, 18513–18529.

Zhong, H., et al. (2016). Content-driven detection of cyberbullying on the instagram social network. *IJCAI, 16*, 3952–3958.